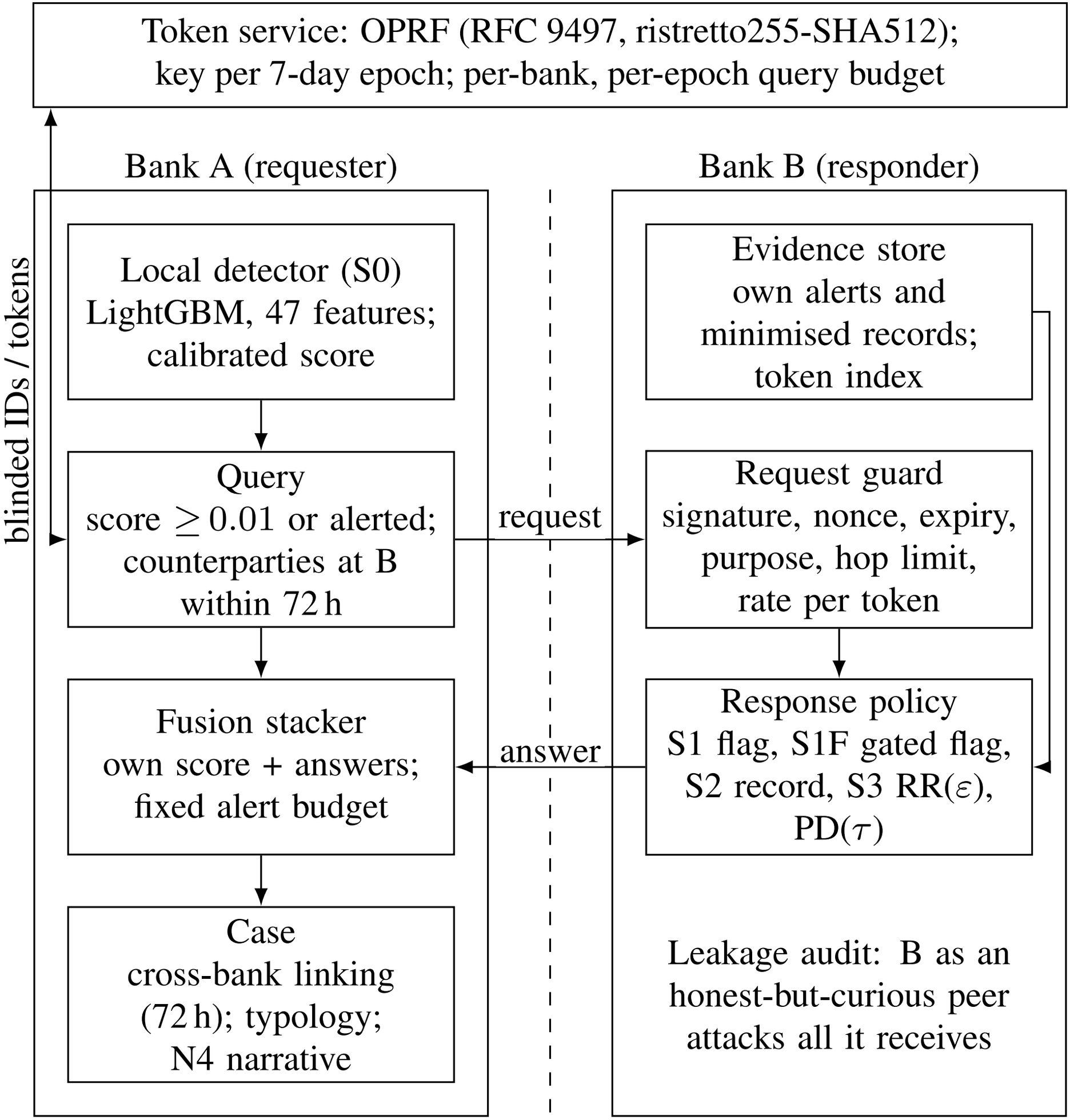
Token service: OPRF (RFC 9497, ristretto255-SHA512);
key per 7-day epoch; per-bank, per-epoch query budget
blinded IDs / tokens
Bank A (requester)
Bank B (responder)
Local detector (S0)
LightGBM, 47 features;
calibrated score
Evidence store
own alerts and
minimised records;
token index
Query
score ≥ 0.01 or alerted;
counterparties at B
within 72 h
request
Request guard
signature, nonce, expiry,
purpose, hop limit,
rate per token
Fusion stacker
own score + answers;
fixed alert budget
answer
Response policy
S1 flag, S1F gated flag,
S2 record, S3 RR(ε),
PD(τ)
Case
cross-bank linking
(72 h); typology;
N4 narrative
Leakage audit: B as an
honest-but-curious peer
attacks all it receives